RAG Learning

T5 — Thermodynamics, Acids and Bases

5.1 Born-Haber Cycles

Born-Haber cycle: An enthalpy cycle (application of Hess's law) used to calculate lattice enthalpy indirectly by combining measurable enthalpy changes: atomisation enthalpies, ionisation energies, electron affinities, and the enthalpy of formation.

Key Enthalpy Terms

Term	Symbol	Definition (per mole, standard conditions)	Sign
Enthalpy of formation	$\Delta_f H°$	Formation of 1 mol of compound from its elements in standard states	Usually negative
Enthalpy of atomisation	$\Delta_{at} H°$	Formation of 1 mol of gaseous atoms from the element in its standard state	Always positive (endothermic)
Ionisation energy	IE	Removal of 1 mol of electrons from 1 mol of gaseous atoms (or ions)	Always positive (endothermic)
Electron affinity (1st)	EA_1	Addition of 1 mol of electrons to 1 mol of gaseous atoms	Usually negative (exothermic)
Electron affinity (2nd)	EA_2	Addition of a second electron to a gaseous 1- ion	Always positive (endothermic — repulsion)
Lattice enthalpy (formation)	$\Delta_{LE} H°$	Formation of 1 mol of ionic lattice from gaseous ions	Always negative (exothermic)

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Lattice enthalpy (dissociation)	$\Delta_{LE}H°$	Breaking of 1 mol of ionic lattice into gaseous ions	Always positive (endothermic)

Exam Tip: Edexcel typically uses the exothermic definition of lattice enthalpy (formation of lattice from gaseous ions, negative value). Always check the question wording. The magnitude of lattice enthalpy is the same for both definitions; only the sign differs.

Constructing a Born-Haber Cycle for NaCl

The cycle connects elements in standard states to the ionic compound via two routes:

Direct route: $Na(s) + ½Cl_2(g) \rightarrow NaCl(s)$ [$\Delta_f H° = -411$ kJ mol^{-1}]

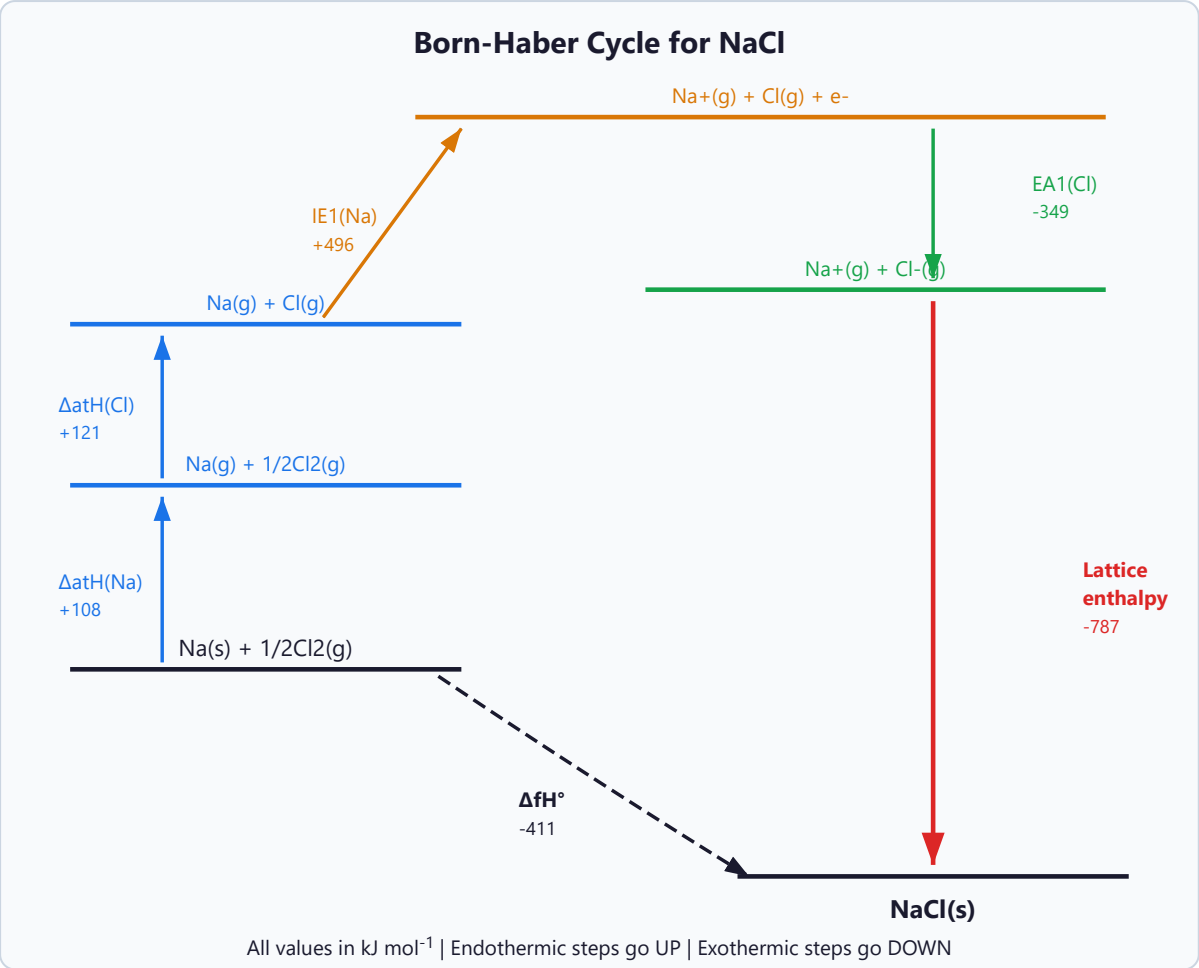
Indirect route (via gaseous ions):

1. Atomise Na: $Na(s) \rightarrow Na(g)$ [$\Delta_{at}H° = +108$ kJ mol^{-1}]
2. Atomise Cl: $½Cl_2(g) \rightarrow Cl(g)$ [$\Delta_{at}H° = +121$ kJ mol^{-1}]
3. Ionise Na: $Na(g) \rightarrow Na^+(g) + e^-$ [$IE_1 = +496$ kJ mol^{-1}]
4. Add electron to Cl: $Cl(g) + e^- \rightarrow Cl^-(g)$ [$EA_1 = -349$ kJ mol^{-1}]
5. Form lattice: $Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$ [$\Delta_{LE}H° = ?$]

By Hess's law: $\Delta_f H° = \Delta_{at}H°(Na) + \Delta_{at}H°(Cl) + IE_1 + EA_1 + \Delta_{LE}H°$

Rearranging: $\Delta_{LE}H° = \Delta_f H° - \Delta_{at}H°(Na) - \Delta_{at}H°(Cl) - IE_1 - EA_1$

$\Delta_{LE}H° = -411 - 108 - 121 - 496 - (-349) = -411 - 108 - 121 - 496 + 349 =$ **-787 kJ mol^{-1}**



Factors Affecting Lattice Enthalpy

Lattice enthalpy becomes more exothermic (larger magnitude) with:

- **Smaller ionic radii:** Ions closer together → stronger electrostatic attraction (Coulomb's law: $F \propto 1/r^2$)
- **Higher ionic charges:** Greater charges → stronger attraction ($F \propto q_1 q_2$)

Example: MgO has a much larger lattice enthalpy (-3850 kJ mol^{-1}) than NaCl (-787 kJ mol^{-1}) because Mg^{2+}/O^{2-} have double the charges and smaller radii than Na^+/Cl^-.

Theoretical vs Experimental Lattice Enthalpies

Theoretical lattice enthalpies (calculated from a perfect ionic model) can be compared with experimental values (from Born-Haber cycles). Good agreement suggests purely ionic bonding. A significant discrepancy (experimental more exothermic than theoretical) indicates covalent character in the bonding — this occurs when the cation has high charge density and polarises the anion's electron cloud (Fajans' rules).

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Exam Tip: If asked why the experimental lattice enthalpy of AgI is much more exothermic than the theoretical value, explain that Ag^+ has a high polarising power (high charge density) and I^- is easily polarised (large, diffuse electron cloud). The resulting covalent character provides additional bonding energy, making the actual lattice enthalpy larger than predicted by a purely ionic model.

5.2 Enthalpies of Hydration and Solution

Enthalpy of hydration ($\Delta_{hyd}H°$): The enthalpy change when 1 mol of gaseous ions is dissolved in sufficient water to form an infinitely dilute solution. Always exothermic (energy released as ion-dipole bonds form between ions and water molecules).

Enthalpy of solution ($\Delta_{sol}H°$): The enthalpy change when 1 mol of an ionic solid dissolves in sufficient water to form an infinitely dilute solution. Can be exothermic or endothermic.

Dissolving involves two processes: (1) breaking the ionic lattice (endothermic, requires energy equal to the lattice dissociation enthalpy), and (2) hydrating the ions (exothermic, releasing hydration enthalpy). The overall enthalpy of solution is the sum:

$$\Delta_{sol}H° = -\Delta_{LE}H°(\text{formation}) + \Delta_{hyd}H°(\text{cation}) + \Delta_{hyd}H°(\text{anion})$$

If the sum of hydration enthalpies exceeds the lattice dissociation enthalpy, $\Delta_{sol}H$ is negative (exothermic dissolution). If not, dissolution is endothermic but may still occur if the entropy increase is sufficient to make ΔG negative.

Factors affecting hydration enthalpy: More exothermic with smaller ionic radius and higher ionic charge (higher charge density creates stronger ion-dipole interactions with water).

5.3 Entropy

Entropy (S): A measure of the disorder or dispersal of energy in a system — specifically, the number of ways energy can be distributed among the particles. Units: $J\ K^{-1}\ mol^{-1}$. The third law of

thermodynamics states that a perfect crystal at absolute zero (0 K) has zero entropy.

Entropy Trends

- **S(gas) >> S(liquid) > S(solid):** Gas particles have the most freedom of movement and the greatest number of energy arrangements.
- Entropy **increases with temperature** (particles have more kinetic energy, more ways to distribute it).
- Entropy increases when the **number of moles of gas increases** in a reaction.
- Entropy increases when a **solid dissolves** in a solvent (particles become more dispersed).
- **More complex molecules** (more atoms, more bonds) have higher entropy than simpler ones (more vibrational modes).

Entropy Change of a Reaction

$$\Delta S_{system} = \Sigma S°(products) - \Sigma S°(reactants)$$

Predicting the sign of ΔS:

- ΔS > 0 (positive): more gas moles in products, solid dissolving, solid → liquid → gas
- ΔS < 0 (negative): fewer gas moles in products, gas → liquid → solid, polymerisation

Example: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$. ΔS is positive because a gas is produced from a solid (increase in disorder).

5.4 Gibbs Free Energy

Gibbs free energy change (ΔG): The thermodynamic quantity that determines whether a reaction is feasible (spontaneous). A reaction is thermodynamically feasible when ΔG < 0 (negative).

$$\Delta G = \Delta H - T\Delta S$$

where ΔH = enthalpy change (kJ mol^{-1}), T = temperature (K), ΔS = entropy change (kJ K^{-1} mol^{-1}). Note: ΔS is often given in J K^{-1} mol^{-1} and must be divided by 1000 before substituting.

Feasibility at Different Temperatures

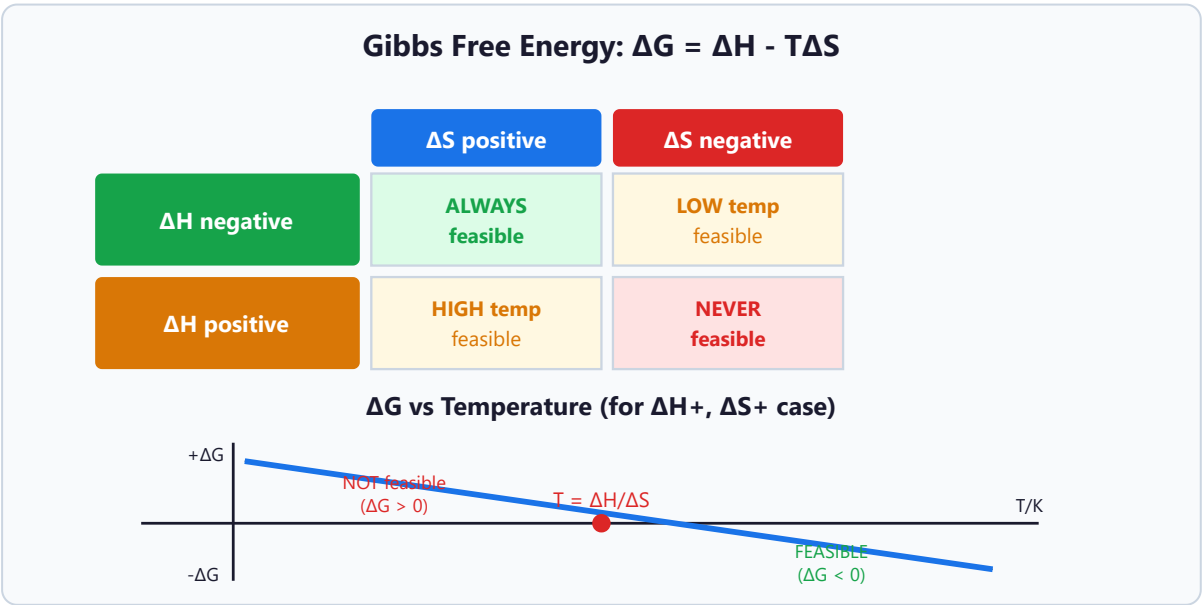
ΔH	ΔS	ΔG	Feasibility
Negative (exo)	Positive (+)	Always negative	Feasible at ALL temperatures
Negative (exo)	Negative (-)	Negative at low T	Feasible at LOW temperatures only
Positive (endo)	Positive (+)	Negative at high T	Feasible at HIGH temperatures only
Positive (endo)	Negative (-)	Always positive	NEVER feasible

The **temperature at which ΔG = 0** (transition between feasible and non-feasible) is found by:

$$T = \Delta H / \Delta S$$

Sample calculation: For $CaCO_3$ decomposition: ΔH = +178 kJ mol^{-1}, ΔS = +0.161 kJ K^{-1} mol^{-1}. T = 178/0.161 = 1106 K (833°C). Below this temperature, ΔG > 0 (not feasible). Above it, ΔG < 0 (feasible).

Exam Tip: Thermodynamic feasibility (ΔG < 0) does NOT guarantee a reaction will occur at an observable rate. Kinetic barriers (high E_a) may prevent it. Example: diamond → graphite has ΔG < 0 at room temperature, but the rate is essentially zero because the activation energy for breaking C-C bonds in diamond is enormous. Always state this limitation when discussing feasibility.



5.5 Acids, Bases and pH

Bronsted-Lowry acid: A proton (H^+) donor.

Bronsted-Lowry base: A proton (H^+) acceptor.

Strong acid: Fully dissociates (ionises) in aqueous solution (e.g. HCl, HNO_3, H_2SO_4).

Weak acid: Partially dissociates in aqueous solution, establishing an equilibrium (e.g. CH_3COOH, H_2CO_3, HF).

Conjugate acid-base pair: Two species that differ by one proton. E.g. CH_3COOH / CH_3COO^-.

pH Calculations

$$pH = -\log_{10}[H^+] \quad \text{and} \quad [H^+] = 10^{-pH}$$

pH of a strong acid: Since strong acids fully dissociate, $[H^+]$ = concentration of the acid (for monobasic acids like HCl). Example: 0.10 mol dm^{-3} HCl has $[H^+]$ = 0.10, pH = -log(0.10) = 1.00.

For dibasic acids like H_2SO_4: $[H^+]$ = 2 × concentration (both protons donated). Example: 0.050 mol dm^{-3} H_2SO_4 has $[H^+]$ = 0.10, pH = 1.00.

5.6 The Ionic Product of Water (K_w)

Water undergoes autoionisation: $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$

$$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}\ mol^2\ dm^{-6} \text{ at } 25°C\ (298\ K)$$

At 25°C, in pure water: $[H^+] = [OH^-] = 1.00 \times 10^{-7}$ mol dm^{-3}, so pH = 7.00.

pH of a strong base: Find $[OH^-]$ from the base concentration, then $[H^+] = K_w/[OH^-]$, then pH = $-\log[H^+]$.

Example: 0.050 mol dm^{-3} NaOH: $[OH^-]$ = 0.050, $[H^+] = 10^{-14}/0.050 = 2.0 \times 10^{-13}$, pH = $-\log(2.0 \times 10^{-13})$ = 12.70.

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Exam Tip: K_w increases with temperature (the forward reaction is endothermic). At temperatures above 25°C, $K_w > 10^{-14}$, so pH of pure water < 7. This does NOT mean the water is acidic — it is still neutral because $[H^+] = [OH^-]$. Neutral pH is only 7.00 at exactly 25°C.

5.7 K_a and pK_a

Acid dissociation constant (K_a): The equilibrium constant for the dissociation of a weak acid in water.

For $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$:

$K_a = [H^+][A^-] / [HA]$ (units: mol dm^{-3})

A larger K_a means a stronger acid (greater degree of dissociation). A smaller pK_a means a stronger acid.

$$pK_a = -\log_{10}K_a \quad K_a = 10^{-pK_a}$$

Calculating pH of a Weak Acid

For a weak acid HA with initial concentration c and K_a known, we make two key assumptions:

1. $[H^+] = [A^-]$ (because they are produced in equal amounts)
2. $[HA]_{equilibrium} \approx c$ (because dissociation is so small that the amount of HA consumed is negligible)

Substituting into K_a:

$$K_a = [H^+]^2 / c \quad \Rightarrow \quad [H^+] = \sqrt{(K_a \times c)} \quad \Rightarrow \quad pH = -\log[H^+]$$

Sample calculation: CH_3COOH, c = 0.10 mol dm^{-3}, $K_a = 1.74 \times 10^{-5}$ mol dm^{-3}.

$[H^+] = \sqrt{(1.74 \times 10^{-5} \times 0.10)} = \sqrt{(1.74 \times 10^{-6})} = 1.32 \times 10^{-3}$ mol dm^{-3}. pH = $-\log(1.32 \times 10^{-3}) = 2.88$.

Exam Tip: State BOTH assumptions explicitly in calculations for full marks. If you need to find K_a from pH, work backwards: $[H^+] = 10^{-pH}$, then $K_a = [H^+]^2/c$. These assumptions are only valid when less

than ~5% of the acid dissociates.

5.8 Buffer Solutions

Buffer solution: A solution that resists changes in pH when small amounts of acid or base are added. An acidic buffer consists of a **weak acid** and its **conjugate base** (usually added as the sodium/potassium salt).

How buffers work: The buffer contains a large reservoir of both HA (weak acid) and A^- (conjugate base):

- **When H^+ is added:** $A^- + H^+ \rightarrow HA$. The conjugate base removes the added H^+, forming more undissociated acid. $[H^+]$ barely changes, so pH is maintained.
- **When OH^- is added:** $HA + OH^- \rightarrow A^- + H_2O$. The weak acid neutralises the added OH^-. $[H^+]$ barely changes.

Making a buffer:

- Method 1: Mix a weak acid with its sodium/potassium salt (e.g. $CH_3COOH + CH_3COONa$)
- Method 2: Partially neutralise a weak acid with a strong base (e.g. add NaOH to excess CH_3COOH). The unreacted acid provides HA, and the salt formed provides A^-.

Calculating Buffer pH

$$[H^+] = K_a \times [HA] / [A^-]$$

$$pH = pK_a + \log_{10}([A^-] / [HA]) \quad \text{(Henderson-Hasselbalch equation)}$$

Sample calculation: Buffer made from 0.20 mol dm^{-3} CH_3COOH and 0.10 mol dm^{-3} CH_3COONa. $K_a = 1.74 \times 10^{-5}$.

$[H^+] = 1.74 \times 10^{-5} \times 0.20/0.10 = 3.48 \times 10^{-5}$. pH = $-\log(3.48 \times 10^{-5}) = 4.46$.

Note: When [HA] = [A^-], then [H^+] = K_a and pH = pK_a. This occurs at the half-equivalence point in a weak acid-strong base titration.

Buffer Action: Resisting pH Change

Acidic buffer: weak acid (HA) + conjugate base (A^-)

$$HA \rightleftharpoons H^+ + A^-$$

H^+ added (acid):

$A^- + H^+ \rightarrow HA$

The conjugate base A^- removes the added H^+ by forming HA

→ [H^+] barely changes
→ pH maintained

OH^- added (base):

$HA + OH^- \rightarrow A^- + H_2O$

The weak acid HA neutralises the added OH^-

→ [H^+] barely changes
→ pH maintained

Buffer capacity depends on concentrations of HA and A^- (higher = better buffering)

5.9 Titration Curves and Indicators

Shapes of pH Titration Curves

Combination	Start pH	Equivalence pH	Suitable Indicator
Strong acid + Strong base	Low (~1)	7.0	Any (methyl orange or phenolphthalein)
Strong acid + Weak base	Low (~1)	< 7 (acidic salt)	Methyl orange (range 3.1-4.4)
Weak acid + Strong base	Higher (~3)	> 7 (basic salt)	Phenolphthalein (range 8.3-10.0)
Weak acid + Weak base	Higher (~3)	≈ 7	No suitable indicator (no sharp endpoint)

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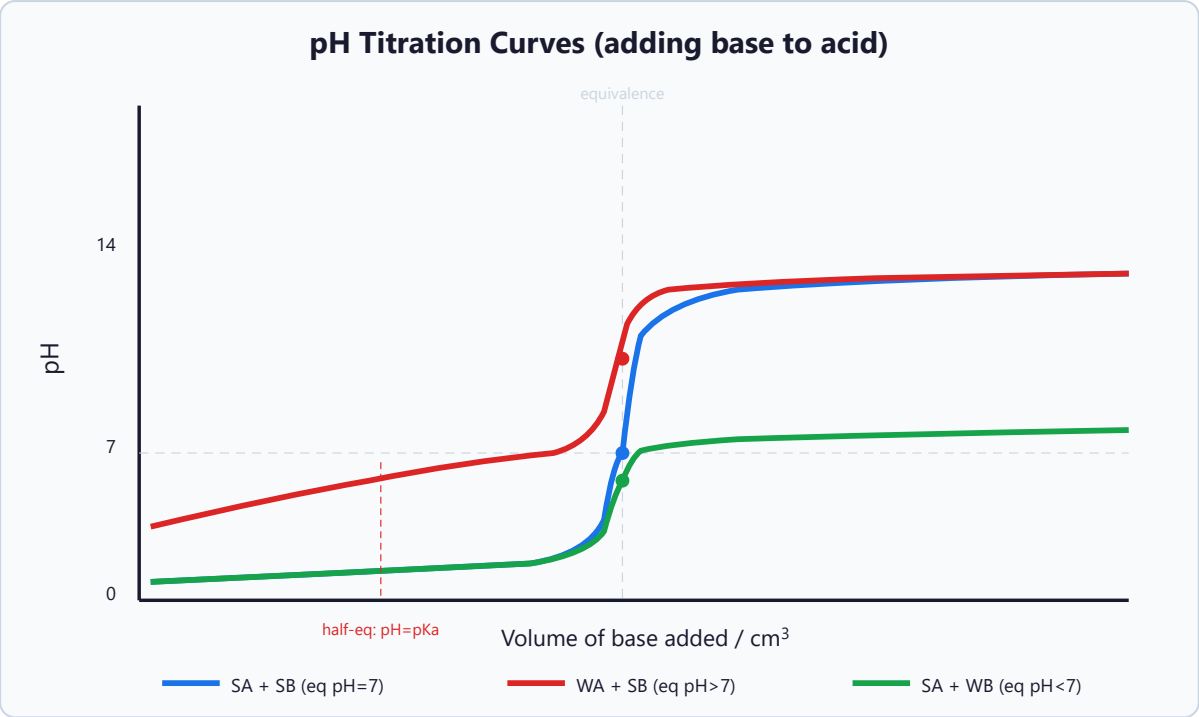

Key Features of a Weak Acid + Strong Base Curve

- **Initial pH** is higher than for a strong acid (only partially dissociated, so fewer H^+ ions)
- **Buffer region:** A relatively flat section before equivalence where pH changes slowly. The solution acts as a natural buffer (mixture of HA and A^-)
- **Half-equivalence point:** Halfway to equivalence, exactly half the acid has been neutralised, so [HA] = $[A^-]$ and pH = pK_a. This allows K_a to be determined experimentally.
- **Equivalence point pH > 7:** The salt formed (NaA) contains the conjugate base A^-, which hydrolyses slightly: $A^- + H_2O \rightleftharpoons HA + OH^-$, making the solution basic.
- **Steep section** around equivalence is shorter than for strong acid-strong base.

Indicators

Indicator: A weak acid (HIn) that changes colour depending on pH. In acid: HIn predominates (one colour). In base: In^- predominates (different colour). The colour change occurs over a range of about 2 pH units centred on pK_{in}.

For a titration, the indicator must have its pK_{in} within the steep (near-vertical) section of the titration curve. This ensures a sharp colour change at the equivalence point.

Indicator	pK_{in}	Colour (acid)	Colour (base)	pH range	Use with
Methyl orange	3.7	Red	Yellow	3.1-4.4	Strong acid + weak/strong base
Phenolphthalein	9.3	Colourless	Pink	8.3-10.0	Weak acid + strong base

Exam Tip: For weak acid + weak base titrations, there is no steep vertical section on the curve, so no indicator gives a sharp endpoint. Use a pH meter instead. At the half-equivalence point, pH = pK_a — this is a very commonly examined relationship. You can read pK_a directly from a titration curve at the point where half the base has been added.

5.10 Enthalpy of Neutralisation

The standard enthalpy of neutralisation is the enthalpy change when an acid and a base react to form one mole of water under standard conditions.

Strong acid + strong base: $\Delta_{neut}H° \approx -57.1$ kJ mol^{-1} (consistent value because both fully dissociate; the reaction is simply $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$).

Weak acid or weak base involved: $\Delta_{neut}H°$ is less exothermic (less negative) because some energy is used to dissociate the weak acid/base before neutralisation can occur. For example, CH_3COOH + NaOH gives $\Delta_{neut}H \approx -55.2$ kJ mol^{-1}.

Exam Tip: If asked why the enthalpy of neutralisation for a weak acid-strong base reaction is less exothermic than expected, explain that energy is required to complete the dissociation of the weak acid ($HA \rightarrow H^+ + A^-$) before neutralisation occurs, reducing the overall exothermic value.